

Gefördert durch:



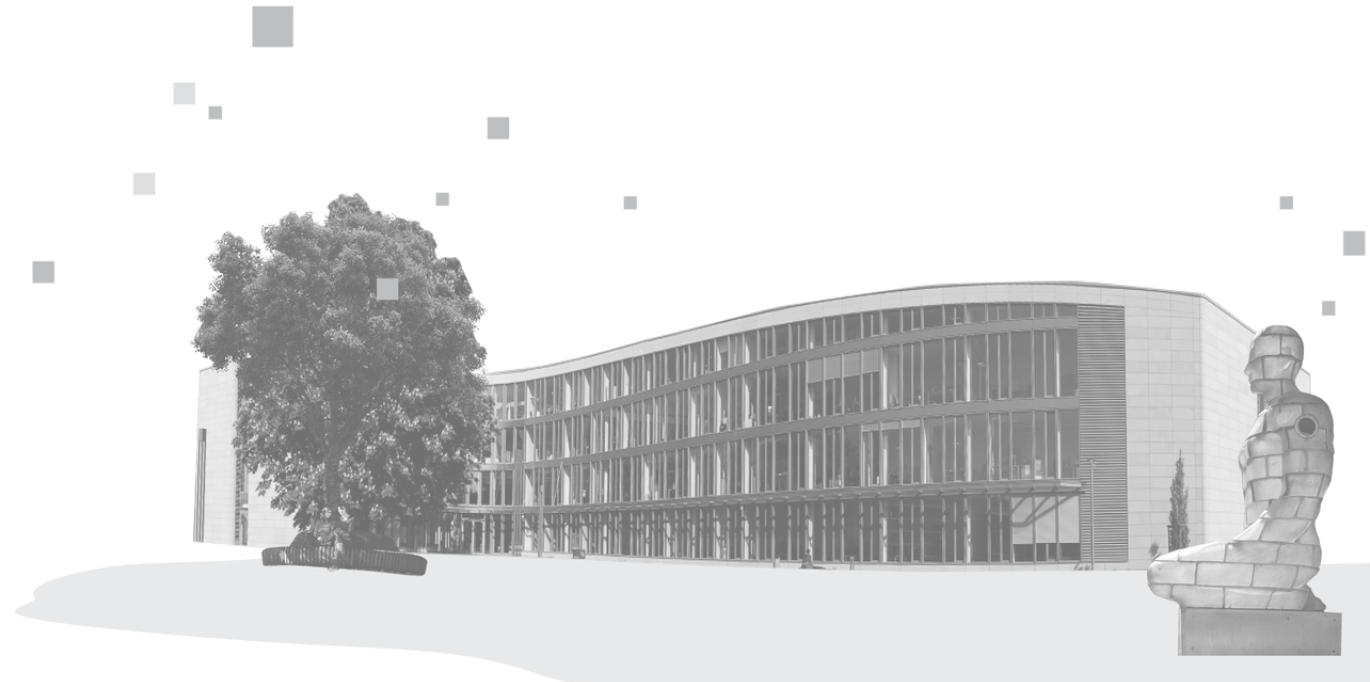
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Workshop How to use our HPC Cluster

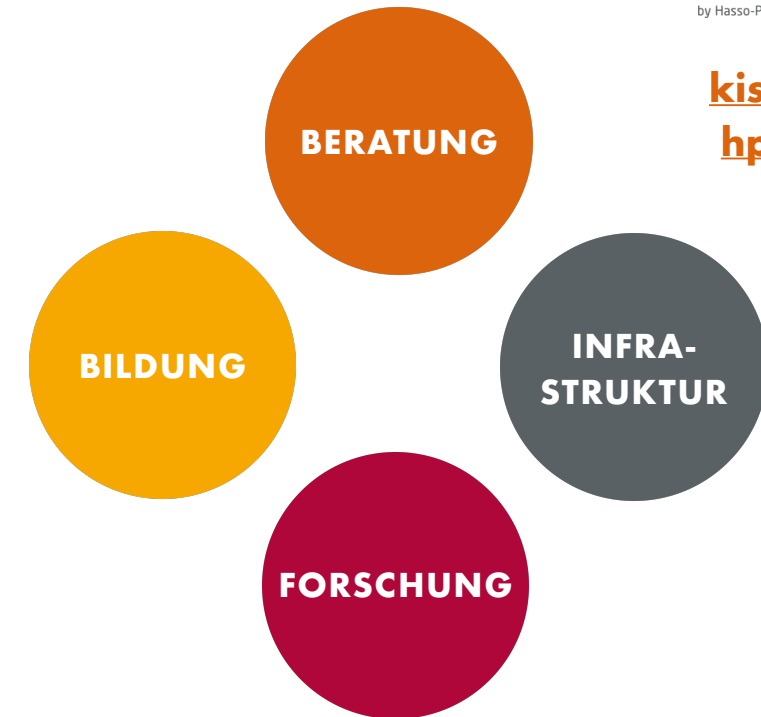
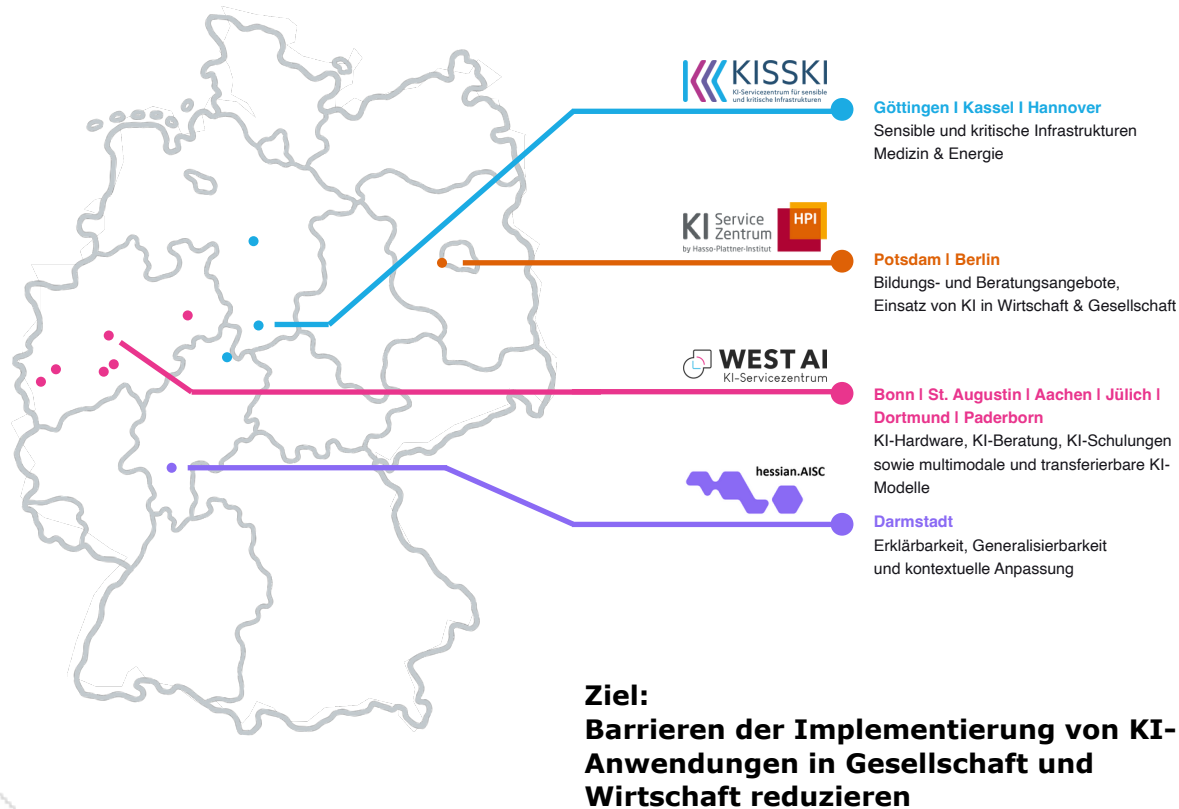
Felix Boelter and David Goll

**Design IT.
Create Knowledge.**

www.hpi.de



KI-Servicezentrum Berlin-Brandenburg



Hasso-Plattner-Institut gGmbH

- Bildet mit **Universität Potsdam** die **Digital Engineering Fakultät**
- Vereint **Forschung, Lehre** mit den Vorteilen einer **privat finanzierten, gebührenfreien Institution**
- Besteht aus Einrichtungen wie der **E-School**, der **D-School**,
- dem **Mittelstandsdigitalzentrum** und dem **KI-Servicezentrum**

Gefördert durch:



Newsletter

BILDUNG

Talks



tele-task.de/series/1463

- Gastvorträge zu Forschung und Innovation

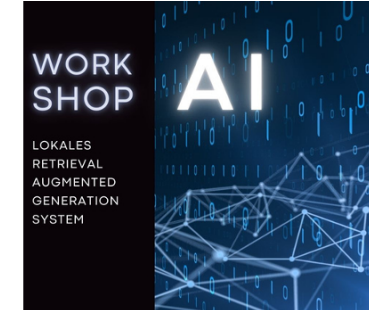


Workshops



aimaker.community

- Praxisnahe Themen
- Beispielthemen: Speech2summary, Docker für ML, semantische Suche

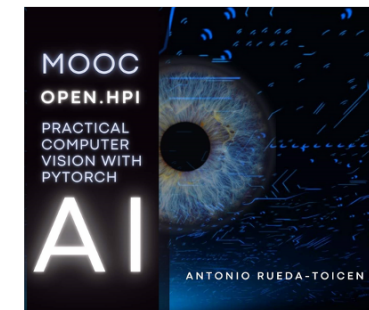


MOOCs



open.hpi.de/channels/ai-service-center

- ChatGPT: Was bedeutet generative KI für unsere Gesellschaft?
- Profitable KI
- KI Biases verstehen und vermeiden



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Sprechstunde buchen

BERATUNG

KI-Sprechstunde

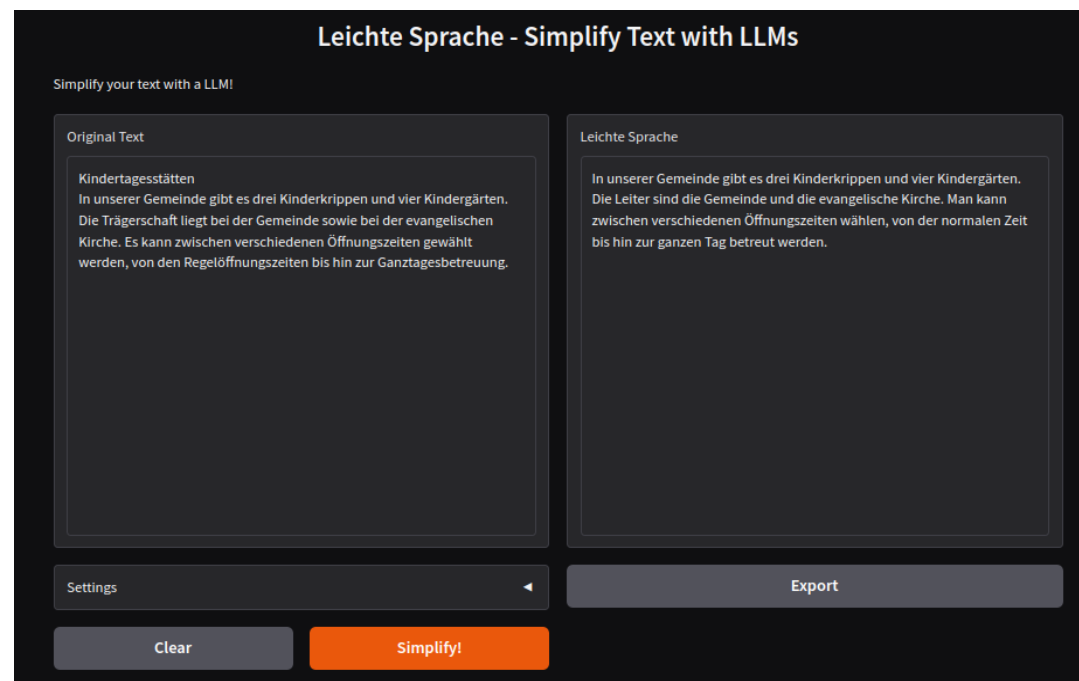
- Beantwortung von Fragen:
 - zu KI-Infrastruktur
 - zu KI-Modellen & Frameworks

KI-Pilotprojekte

- Co-Entwicklung eines Prototyps
- Bewerbung alle drei Monate
- Auswahlkriterien z.B. KI-Reife, Gemeinwohl
- Veröffentlichung der Ergebnisse

Kooperationen

- Gemeinsam organisierte Netzwerktreffen



Bisherige KI-Pilotprojekte

- Generierung Mathematik-Problemen
- Leichte Sprache
- Generierung von Upcycling Vorschlägen
- Reduzierung von Food Waste
- Datierung mittels Handschrift



Jetzt bewerben!

github.com/aihpi/leichte-sprache

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Zugangsanfrage
aisc.hpi.de



Bundesministerium
für Bildung
und Forschung

INFRASTRUKTUR



- Zugang **kostenfrei**
- Kein Produktionsbetrieb
 - Daten sollten **anonymisiert** oder **synthetisiert** sein
 - kein **Hosting** von Produkten
- **Reporting & Veröffentlichung** durch Nutzende
- **Altrechte** bleiben bei Nutzenden
- **Neurechte** bleiben bei Nutzenden
 - Einräumen von Nutzungsrechten für Forschung und Lehre

Training

- 64 NVIDIA H100 GPU

Edge

- ARMv8 CPU
- NVIDIA Jetson AGX Module

Inferenz

- 40 NVIDIA A30 GPU

Neuromorph

- 288 SpiNNaker2 Chips

ARM Server

- Ampere Altra Max M128-30 CPU
- 2 x NVIDIA L40 GPUs

Speicher

- 1.5 PB NVRAM

Netzwerk

- 400 Gb/s Infiniband
- 200 Gb/s Ethernet

GPU Server

- AMD Epyc CPU
- 8 x NVIDIA L40S GPU

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AGENDA

Workshop Structure

Part	Topic	Duration
I	Introduction and Motivation	10 min
I	Interactive: Cluster access setup	20 min
I	Slides: partitions, nodes, SLURM, Interactive vs. Batch	10 min
I	Interactive: VSCode Installation	5 min
<i>Break</i>		
II	Interactive SLURM workflow	10-20 min
II	Slides: Basics of CPUs/GPUs and fairshare	10-20 min
II	Interactive: Cloning of repository	5 min
II	Interactive: Batch scripts (01 - 08)	20 min
II	Conclusion and Feedback	5 min

Motivation

Scale: Train models that don't fit on consumer hardware

Speed: Powerful GPUs, run experiments in parallel

Freedom: Submit and disconnect

Collaboration: Shared environment across your team

Cost: Free for AISC projects



Documentation

Please make yourself familiar with the HPC documentation!

If you have questions or need assistance, consult our [FAQ](#) or reach out to us:

For technical issues:
aisc-helpdesk@hpi.de

For organizational issues:
ki-servicezentrum@hpi.de

Hasso
Plattner
Institut

Scientific Compute Documentation

 Search

 SC Wiki (HPI GitLab)

Home

FAQ

Cluster (HPC/Slurm)

Cloud (VMs/Services)

AISC

Linux Basics

SSH

VPN

Cluster (HPC/Slurm)

SLURM (Scheduler)

Slurm Basics

Job Examples

Our Customizations

Scheduling and Priorities

Environmental Variables

Slurm REST API

Processes

Access to the HPC Cluster

Group Shares / Project Storage

Password Reset

Resources

Partitions

Hardware

Naming Convention

Node Features

Login Nodes

Run Nodes

Containerization

Enroot

Pyxis (Slurm Container Plugin)

Using Slurm within a container

Storage / Data Transfer

Storage Overview

Quotas / Limits

Scratch Space

Data Transfer

Software Installation

Python / Conda

Sudo / Superuser Permissions

Visual Studio Code

Jupyter Lab / Notebook

Matlab

PyCharm

R Language / RStudio

Cluster (HPC/Slurm)

The cluster offers powerful shared **resources** (CPU and GPU) for your computation jobs. You can run them interactively (in a shell like on your computer) or spawn a bunch of them at once with a batch file.

See **Slurm Basics** for the general concepts and more details on job management.

 **Tip**

No access to the HPC cluster yet? Please check the **access documentation** first.

Quick Start

- Request access via sc-helpdesk@hpi.de (if you don't have it yet)
- Join the Slack channel [#sc-announce](#) for general announcements (e.g. maintenance, etc.), as well as [#sc-discuss](#) for questions and discussion about the cluster with other users.
- Connect to the **Scientific Compute VPN** (This is not the same as the HPI VPN!)
- Connect via SSH (name must be lowercase): `ssh {first.lastname}@hpc.sci.hpi.de`
- Pick a partition: e.g. `cpu-interactive` (also see **Partitions**)
- Pick a account you are allowed to use: e.g. `sci-demo-account` (run `saccount` to see your accounts)
- Run the job, e.g. `srun --partition=cpu-interactive --account=sci-demo-account --cpus-per-task=2 --gpus=0 --mem=1G --time=00:30:00 --pty bash`
- Run your job: `python3 myscript.py` (also see **Job Examples**)
- Leave the node: `exit` or `^Ctrl + D`

 **Attention**

Do not run anything on the access nodes, but on the compute nodes allocated to you via Slurm. The access nodes block long running jobs and may be restarted monthly.

Do not run vscode on the login nodes! Please see our guide to [using vscode on the cluster](#) or use a **Run Node** instead.

 **Tip**

See the Software Installation section in the menu on the left for instructions on how to install common data science tools like **R**, **Jupyter Notebooks**, or **Conda**.

Table of contents

Quick Start

SSH Public Key Authentication

SSH Host Key Verification

Initial Setup

<https://github.com/aihpi/workshop-slurm>

Getting Started

Cluster Access Setup

You can find all of this information in the [Scientific Compute Documentation](#). Please note that the Docs provide information for the [general HPI HPC](#), of which the [AISC infrastructure](#) is a part of. We outline the important steps below:

1. Get access via the AISC portal by submitting a [proposal](#):
2. When your account is created, you should receive a .zip file via mail, containing your username, initial password, and, crucially, an .ovpn file.
3. Set your password via the steps explained in the .zip file
4. Set up a [VPN connection](#) and keep it running
5. Set up a SSH connection with the cluster by following the steps outlined to the left in [SSH guide](#)
6. With your VPN connection **active**, connect to the cluster via SSH: `ssh user.name@hpc.sci.hpi.de`

Optional: Instead of typing the full address every time, create or edit `~/.ssh/config` and add:

```
Host hpi-hpc
  HostName hpc.sci.hpi.de
  User user.name
```



After saving, you can connect with just `ssh hpi-hpc`. You can replace `hpi-hpc` with any alias you prefer.

7. Once you are connected, please take some time to familiarise yourself with the basic concepts of [SLURM](#), [jobs](#) (especially SBatch files), [partitions](#) (only the aisc-... partitions are relevant for you), and [run nodes](#), `nvidia-smi`, ...

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Cluster Basics



Nodes

Cluster Basics

- Cluster = many computers (nodes) connected over a network
- Each node has its own CPUs, RAM, (sometimes) GPUs, ...
- Three types of nodes on our cluster
 1. Login node (lx01)
 - Entry point, like a lobby
 - Used only for file management, submitting jobs, ...
 2. Run node (rx01/rx02)
 - Lightweight interactive work, VSCode, ...
 - Shared usage. Per-user limit: 4 CPUs, 8GB RAM
 3. Compute node (gx[07-14], gx15v1, gx17v1, ...)
 - Any computation (also interactive), ML training, data processing, ...
 - Exclusive usage. Via SLURM only!

What is SLURM?

- Job scheduler: manages who gets what resources and when
- You request resources through SLURM
- Jobs are prioritised based on **availability** and **fairness**

SLURM Basics

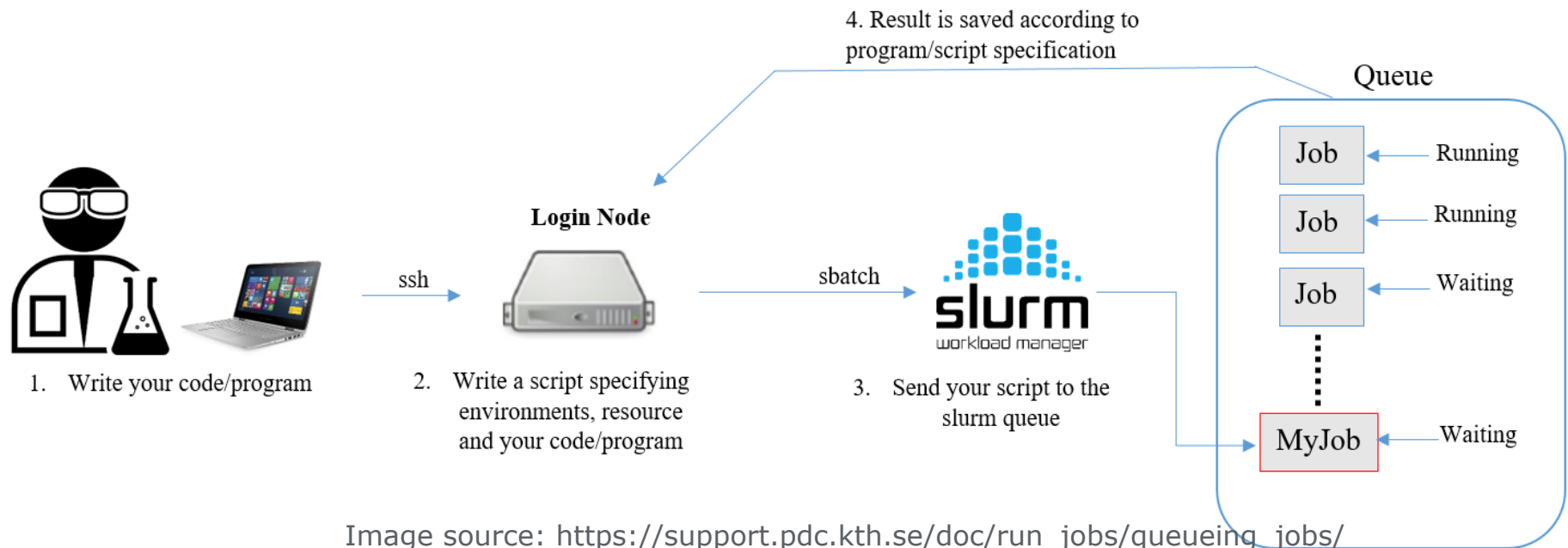


Image source: https://support.pdc.kth.se/doc/run_jobs/queueing_jobs/

Key Concepts

- Partition: group of nodes with shared rules (time limits, hardware type)

Partition	Time Limit	Use Case
pot-hpi-aisc-shortrun	1h	Quick interactive tests (no sbatch)
pot-hpi-aisc-interactive	8h	Interactive terminal, testing
pot-hpi-aisc-batch	7 days	Long training runs (only accessible via sbatch jobs)

- Account: Your project's billing code. In our case it is always `aisc`.
—> You can find out which accounts you have via `saccount`
- Quality of Service (QOS): Used to prioritize certain jobs. We use `aisc`.
—> You will likely not need to use this

SLURM Commands

Command	Functionality
sinfo	Overview of the CPU and GPU usage of the cluster (wrapped for prettier output)
sfree	View available resources to select them for a job
squeue	View submitted jobs and their state
sjob	Show details about a certain job
snode	Show details about a certain node
salloc	Obtain a Slurm job allocation (a set of nodes), execute a command, and then release the allocation when the command is finished
srun	Execute command on the resources assigned to a job
saccount	Show the accounts you are allowed to use
sbatch	Submit a batch script to Slurm
swatch	View current resource usage for the active job

<https://docs.sc.hpi.de/cluster/SLURM/Basics/>

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Two ways to work on compute nodes

	Interactive		Batch
Command	srun	salloc/sblock	sbatch
Output	Live in terminal	Live in terminal	Written to log file
Best for	Quick commands, sanity checks	Interactive work	Training runs, anything serious

```
[d.goll@lx01:~]$ cat test.py
print("Hello World!")
[d.goll@lx01:~]$ python test.py
This command is not allowed on the login node!
Please run resource-intensive commands in a Slurm job instead to avoid overloading the login node.
HINT: Use 'srun' to start an interactive job on a compute node.
HINT: Use a run node for lightweight interactive tasks.
See https://docs.sc.hpi.de/cluster/Resources/Run-Nodes/
[d.goll@lx01:~]$ srun --account=aisc --partition=aisc-interactive --time=00:01:00 --mem=1G --pty bash
[d.goll@gx15v1:~]$ python test.py
Hello World!
[d.goll@gx15v1:~]$ exit
exit
```

Two ways to work on compute nodes

	Interactive		Batch
Command	srun	salloc/sblock	sbatch
Output	Live in terminal	Live in terminal	Written to log file
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```
[d.goll@lx01:~]$ cat test.py
print("Hello World!")
[d.goll@lx01:~]$ python test.py
This command is not allowed on the login node
Please run resource-intensive commands in a batch job
HINT: Use 'srun' to start an interactive job
HINT: Use a run node for lightweight interactive jobs
See https://docs.sc.hpi.de/cluster/ for more information
[d.goll@lx01:~]$ srun --account=aisc --partition=aisc python test.py
[d.goll@gx15v1:~]$ python test.py
Hello World!
[d.goll@gx15v1:~]$ exit
exit
```

```
[d.goll@lx01:~]$ cat test.sh
#!/bin/bash
#SBATCH --account=aisc
#SBATCH --partition=aisc-batch
#SBATCH --time=00:01:00 # 1 minute
#SBATCH --mem=1G

python test.py
[d.goll@lx01:~]$ sbatch test.sh
Submitted batch job 1848135
[d.goll@lx01:~]$ ls
bin  job.logs  slurm-1848135.out  test.py  test.sh  workshop-slurm
[d.goll@lx01:~]$ cat slurm-1848135.out
Hello World!
```

Where to run things:

- rx01/02 are for lightweight tasks only, not a substitute for compute nodes

Resource usage:

- Always specify expected time/memory usage!

Batch jobs:

- Prefer sbatch over interactive jobs for anything serious

Being a good cluster citizen:

- Don't leave idle interactive sessions running. Release resources you're not using.

Read the Terms of Usage: <https://docs.sc.hpi.de/Terms-of-Usage>

Rules and Best Practices

Gefördert durch:

Setting up a Connection with VSCode

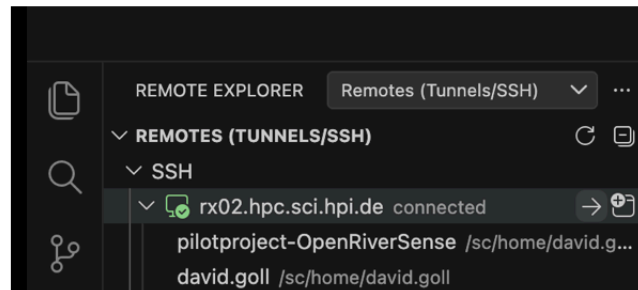
When you log in via `ssh user.name@hpc.sci.hpi.de` you are connected with one of the [login nodes](#), e.g. `1x01`. The login nodes are only intended for submitting jobs to Compute nodes, inspect the cluster status, and manage your files. You cannot execute scripts on the login nodes. *They are also not intended for a connection to VSCode.* For this, please use one of the [run nodes](#).

1. In VSCode, open the Command Palette: Shift + Command + P (Mac) / Ctrl + Shift + P (Windows/Linux).
2. Enter `Remote-SSH: Add New SSH Host...`
3. Enter your username and the run node address in the following format: `user.name@rx01.hpc.sci.hpi.de` / `user.name@rx02.hpc.sci.hpi.de`
4. This will write the following information to your `~/.ssh/config` file:

```
Host rx01.hpc.sci.hpi.de
  HostName rx01.hpc.sci.hpi.de
  User user.name
```



5. Now you should be able to connect easily to the run nodes via the Remote Explorer on the left in VSCode:



VSCode Setup

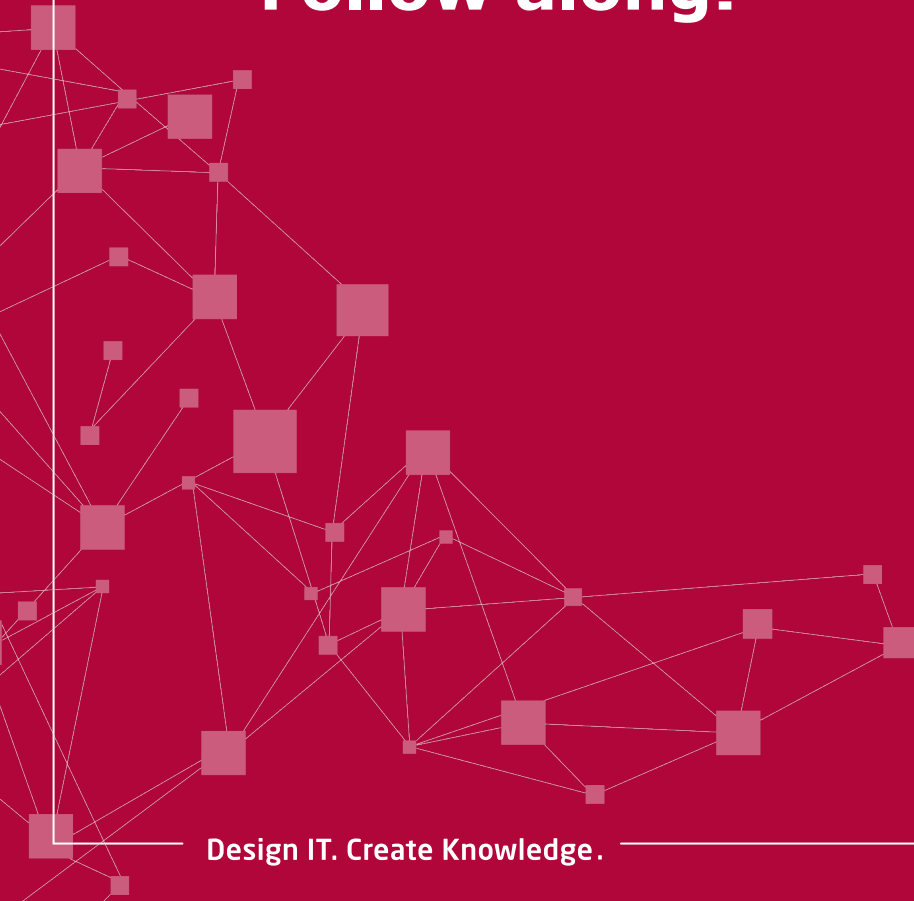
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Short Break

Interactive SLURM

**Using our Cluster with VSCode interactively.
Follow along!**



Part II

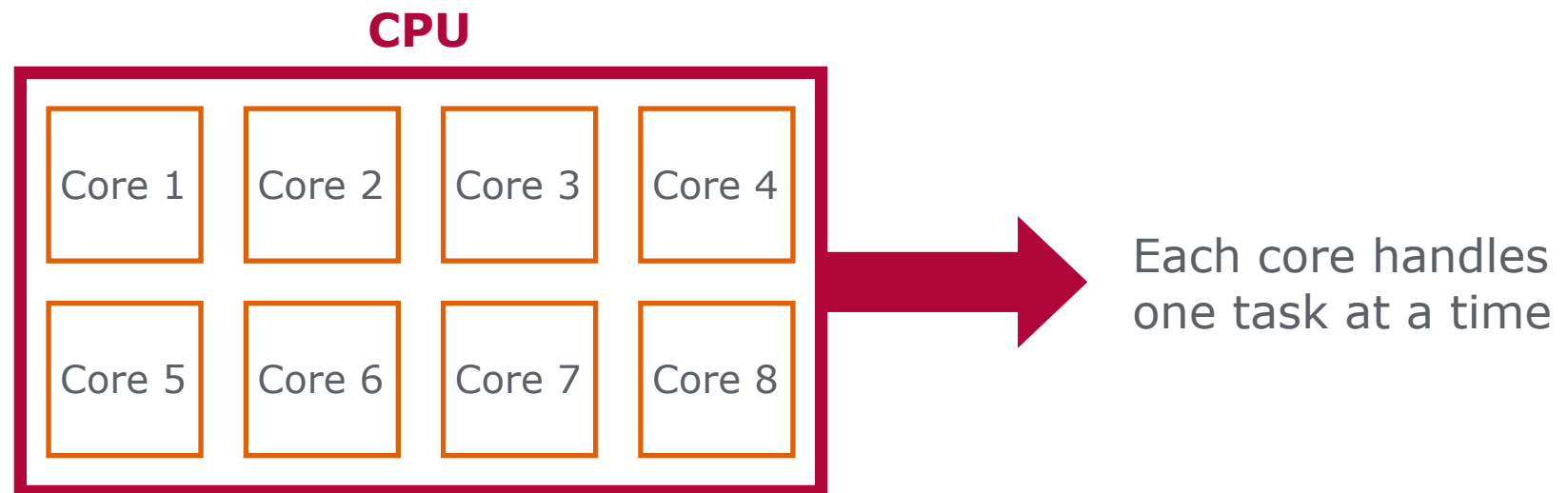
CPUs, GPUs & AI Workloads



CPU

Central Processing Unit

- Few powerful cores, optimized for sequential tasks
- Handles logic, branching, control flow, I/O
- Our H100 nodes:
 - 2x Intel Xeon 8480C, 112 cores (224 threads)

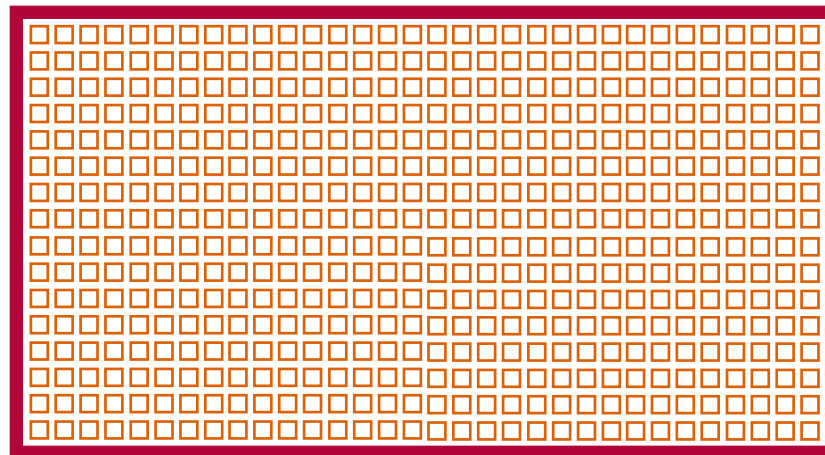


GPU

Graphics Processing Unit

- Originally designed for rendering pixels on screen
- Thousands of small, simple cores working in parallel
- One H100 has 16,896 CUDA cores (vs 112 CPU cores on the same node)
- Key insight: the same parallelism that renders pixels also powers matrix multiplications in neural networks

GPU



Thousands of cores each
doing one simple operation

CPU vs GPU

CPU excels at

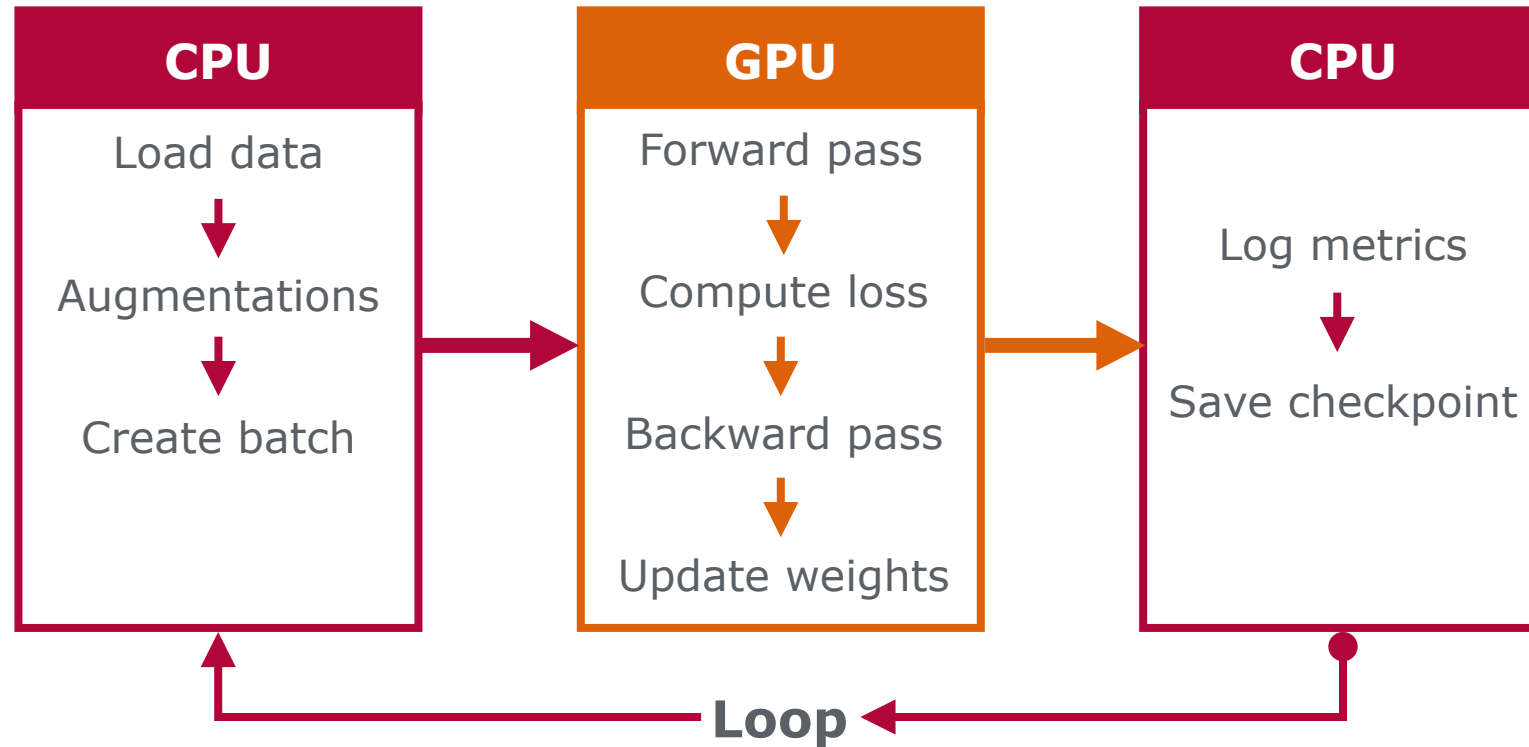
- Data loading and preprocessing (pandas, file I/O)
- Control flow and branching logic
- Classical ML models (sklearn, XGBoost on small data)
- Orchestrating the training pipeline

GPU excels at

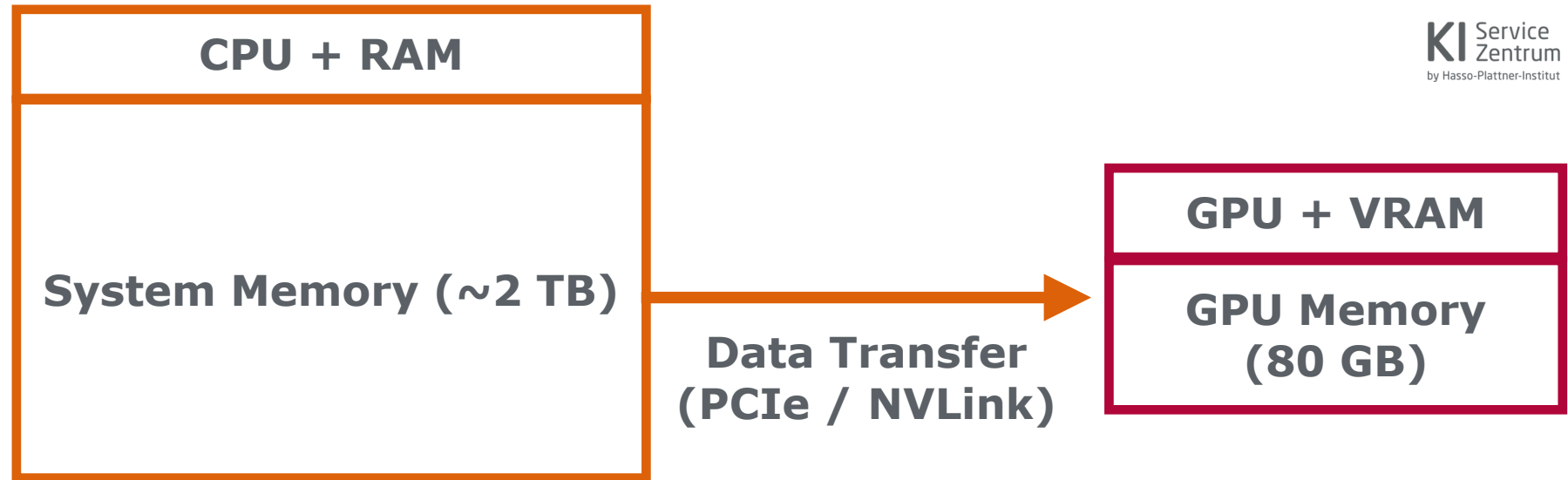
- Matrix multiplications (core of every neural network)
- Model Training and Inference
- Applying the same operation to lots of data in parallel (Single Instruction Multiple Data, **SIMD**)

Gefördert durch:

CPU + GPU



- CPU feeds data, GPU does the math
- CPU too slow → GPU sits idle ("CPU bottleneck")
- Request both in SLURM: **--cpus-per-task** and **--gpus**
- Rule of thumb: 4-8 CPUs per GPU
- Our H100 nodes: up to 14 CPU cores per GPU available



RAM vs VRAM

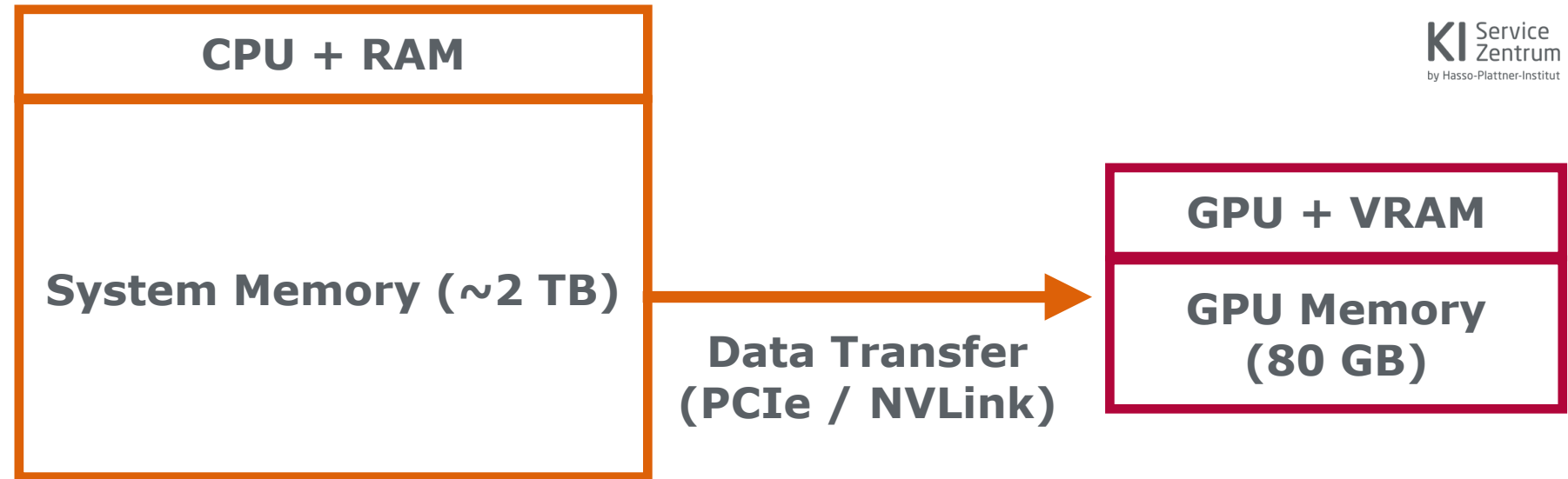
RAM (system memory):

used by the CPU. Holds your dataset, preprocessing pipelines, OS, and everything else. Our H100 nodes have ~2 TB.

VRAM (GPU memory):

dedicated memory on the GPU. Holds model weights, activations, and gradients during training. An H100 has 80 GB.

Gefördert durch:



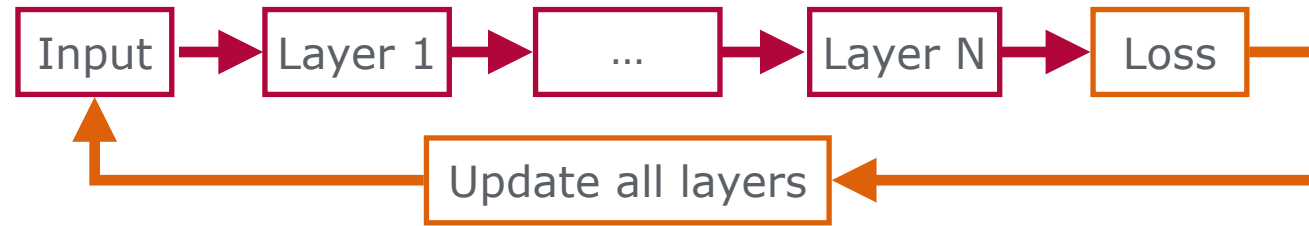
RAM vs VRAM

- 7B parameter model in fp16 $\approx$ 14 GB just for weights. Add activations and gradients $\rightarrow$ easily fills 80 GB
- Out of RAM $\rightarrow$ job gets killed
- Out of VRAM $\rightarrow$ **CUDA out of memory**

Gefördert durch:

Training vs Inference

Training



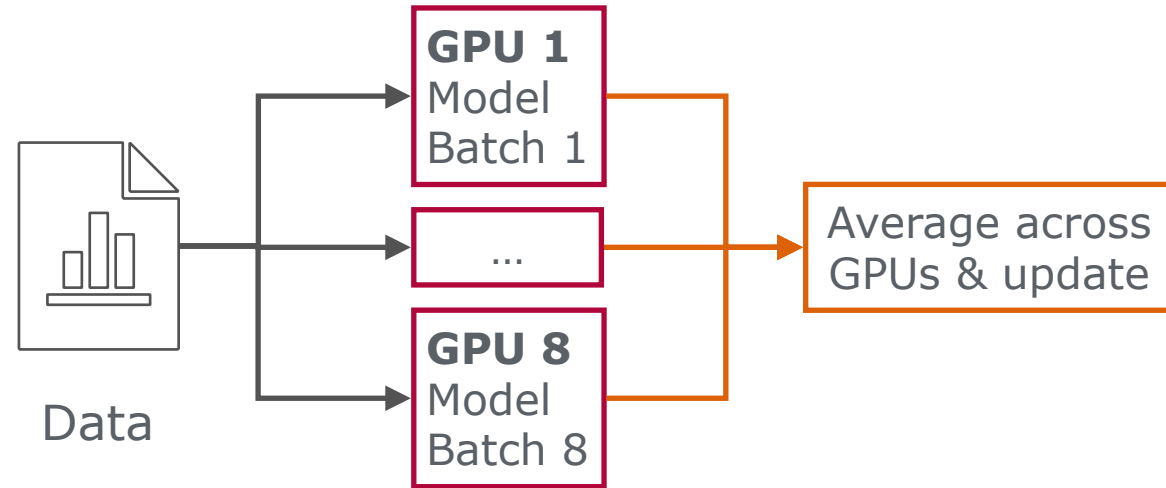
Inference



- Training: forward + backward pass, store activations, update weights. Repeat millions of times.
- Inference: forward pass only. Less VRAM, less compute.
- H100s (80 GB) for training, A30s (24 GB) for inference.
- Our inference infrastructure runs on Kubernetes with ArgoCD.
- Future workshop planned for model deployment.

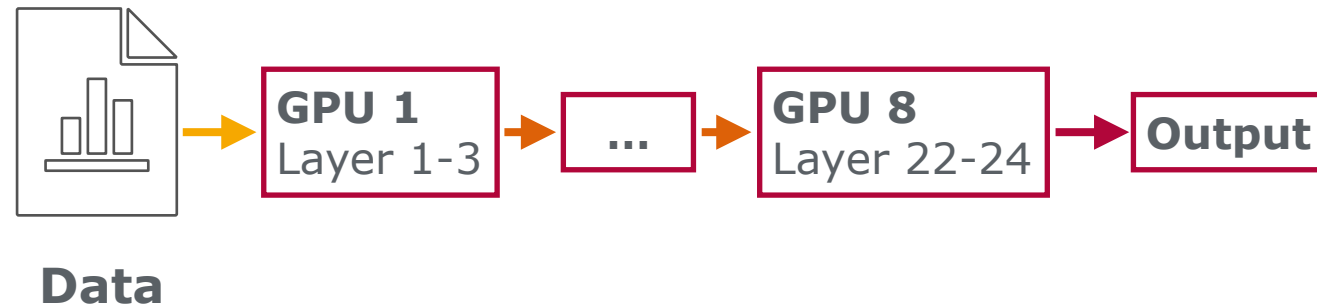
Multi-GPU

Data Parallelism



- Same model copied to every GPU, each gets a different batch
- GPUs compute in parallel, then sync gradients
- Most common approach, scales training speed roughly linearly
- Frameworks: **torchrun**, **Hugging Face Accelerate**, **DeepSpeed**
- This is what we will use in the hands-on exercise

Model Parallelism



Multi-GPU

- Model is split across GPUs, each holds a portion of the layers
- Needed when the model doesn't fit on a single GPU
- Example: 70B LLM in fp16 $\approx$ 140 GB, doesn't fit on one H100 (80 GB)
- More complex to set up, use when data parallelism isn't enough
- More common in inference, where large models need multiple GPUs just to be loaded
- Start with one GPU, scale up when you hit a bottleneck

Gefördert durch:

Our GPUs at a Glance

GPU	Count	VRAM	Best for
H100	64	80 GB	Large model training, multi-GPU jobs
A30	40	24 GB	Inference, smaller training, interactive queues
L40s	8	48 GB	Mixed workloads
L40	2	48 GB	Mixed workloads (ARM server)

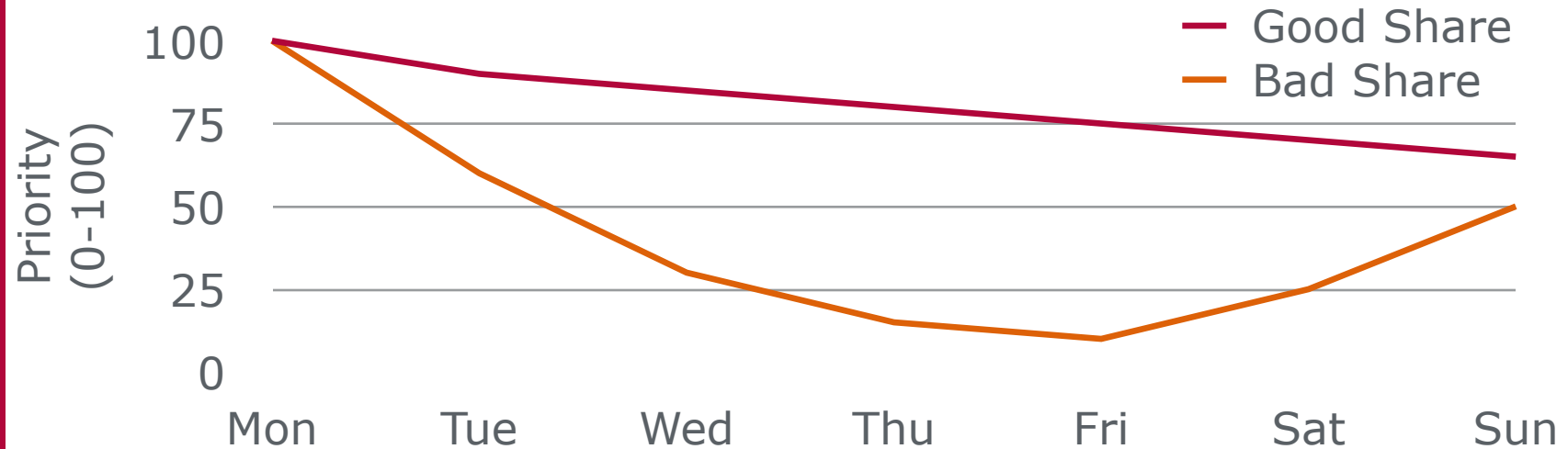
Note: `pot-hpi-aisc-interactive` runs on A30s,
`pot-hpi-aisc-batch` runs mainly on H100s.

H100 Nodes (gx07-gx14)

- 2x Intel Xeon 8480C, 112 cores / 224 threads
- ~2 TB RAM • 8x H100 80 GB per node • 28 TB NVMe
- NDR InfiniBand + 100G Ethernet
→ 14 CPU cores and ~256 GB RAM per GPU

Gefördert durch:

Fairshare



Note: Your priority recovers over time when you're not using resources

- Everyone wants GPUs. SLURM Fairshare decides who goes first.
- Use more → priority drops. Stop using → priority recovers.
- Check your score: `sshare -u $USER`
- Test your pipeline in `pot-hpi-aisc-interactive` first, then submit long runs via `pot-hpi-aisc-batch`.
- **Important:** don't request more than you need, don't leave jobs idle.

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Clone the repository

github.com/aihpi/workshop-slurm

 **workshop-slurm** Private Edit Pins Watch 0

main 1 Branch 0 Tags

Go to file

Add file Code

golldavid Update CHANGELOG with v0.3.0 changes

00_aisc/img	Started working on	
scripts	Add data setup scri	
.gitignore	Add FAQ section to	
.python-version	Rework scripts and	
CHANGELOG.md	Update CHANGELO	
LICENSE	Initial commit	
README.md	Add FAQ section to README and update .gitignore	6 hours ago
TODO.MD	Update TODO list: mark CPUs/GPUs and Fairshare slides a...	last week
pyproject.toml	Add CIFAR-100 single/multi-GPU training scripts (06)	last week
uv.lock	Add CIFAR-100 single/multi-GPU training scripts (06)	last week
workshop-slurm.key	Add Part 2 to Slides	last week

Local

Codespaces

Clone

HTTPS SSH GitHub CLI

Copy URL to clipboard

https://github.com/aihpi/workshop-slurm.g

Clone using the web URL.

Open with GitHub Desktop

Download ZIP

README MIT license

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 **KI Service Zentrum**
by Hasso-Plattner-Institut



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AISC Cluster Workshop: Introduction to SLURM

This repository contains materials for the 2-hour HPC Cluster Workshop by the [AI Service Centre Berlin-Brandenburg \(AISC\)](#). Participants learn how to access the AISC/HPI cluster, understand the basics of SLURM, and run GPU workloads independently.

SBATCH Jobs

**Submit jobs via scripts.
Follow along!**

kisz@hpi.de

hpi.de/kisz

Ihre Meinung ist
uns wichtig!



QR-Code zum Feedback-Formular

